SSGRAM: 3-D Spectral–Spatial Feature Network Enhanced by Graph Attention Map for Hyperspectral Image Classification

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***Abstract*— Convolutional neural networks (CNNs) and graph neural networks (GNNs) are two widely used architectures in hyperspectral image (HSI) classification. Most CNN models tend to heavily rely on neighboring pixels of the target pixel, leading to performance degradation when target pixels differ from their surrounding pixels. To address this limitation, this article introduces a novel approach to HSI classification using a combination of CNNs and GNNs to complement the drawbacks of CNN models. We propose a CNN-based 3-D spectral–spatial feature network (3D-S2FN) to extract features in spectral, spatial, and spectral–spatial spaces, utilizing pyramid squeeze attention to prioritize their importance. Additionally, we introduce a graph attention feature processor (GAFP) module that evaluates the relevance of neighboring HSI pixels to the target pixel, generating a graph attention map (GRAM). This GRAM is applied to the CNN's intermediate features to mitigate neighborhood bias. Unlike conventional GNNs, the proposed GAFP module directly processes the data within a local window, without requiring dimensionality reduction methods that would diminish pixel-level detail. By integrating features from the GAFP and the attention-enhanced 3D-S2FN, the proposed SSGRAM method achieves superior classification performance, as demonstrated through extensive experiments on four publicly available datasets. The model code has been published in https://github.com/BishmoyPaul/SSGRAM**

***Index Terms*— Convolutional neural networks (CNNs), graph neural networks (GNNs), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL imaging techniques can capture detailed geospatial and spectral information from ground objects across hundreds of contiguous high-resolution spectral bands [1], [2]. This spectral information can be analyzed to identify objects. Hyperspectral image (HSI) classification involves classifying ground objects based on the available band information and achieving optimum performance in the automated classification of HSI can benefit numerous fields [3], [4], [5], [6].

HSI classification methods typically take an HSI cube as input and classify the objects based on their spectral information. After the potential of end-to-end deep learning model in HSI classification is demonstrated in [7], different approaches are investigated to improve the model performance, such as increasing model depth [8], using depth-wise convolutions [9], processing both spatial and spectral dimensions [10], using recurrent neural networks for sequential processing [11], [12], utilizing transformers to process long-range dependencies [13] or applying attention mechanisms to the spatial and spectral dimensions [14]. Convolutional neural network (CNN)-based methods are good at modeling local spatial dependencies, and so they have a distinct advantage in processing input HSI cubes containing spatial information relevant to classification. But they can potentially use the information surrounding the central pixel as a shortcut to classify the central pixel. As a result, when the input 3-D HSI cube contains neighbors with different classes, model performance suffers in unseen test cases [15]. Therefore, it is important to consider potential diversity in the surrounding area of the patch for improved performance. To address this issue, [16] utilized graph neural networks (GNNs) in parallel with CNNs for improved classification performance.

While graph-based deep learning models have been employed in HSI classification by themselves [16], [17], [18], [19], [20], CNN- and GNN-based networks exhibit different strengths, and their combination might yield better results over any single type of model, as noted by [16]. Several subsequent studies further explored the integration or fusion of these two networks [18], [19]. However, these methods are reliant on the initial graph generation method for the proper flow of information to graph networks. Existing methods commonly use simple linear iterative clustering (SLIC)-based superpixel generation [16], [17], [21], [22] (introduced in [23]) or CNN-based dimensionality reduction techniques [18], [19] to create an initial graph. SLIC-based approaches segment the input into superpixel regions with HSI pixel mean value, which causes a loss in the discriminativeness of the pixels [24]. As a result, they struggle with modeling individual pixel information, and lead to suboptimal performance. Moreover,

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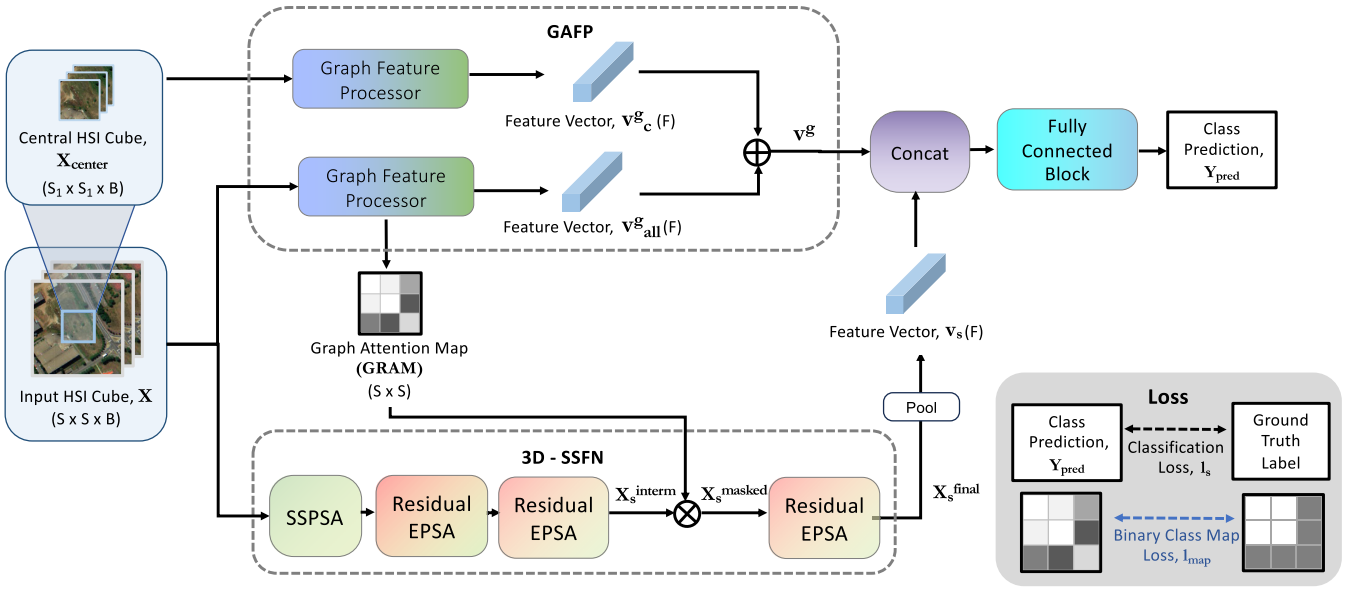


Fig. 1. Graphical overview of the proposed scheme. The components that make up the proposed GAFP and 3D-S2FN module are shown in the figure. The losses used in the proposed method are described in the lower-right corner. The classification loss, denoted with the black dotted line, compares class prediction with ground truth. The blue dotted line indicates the binary class map prediction scheme, where the GRAM is compared with a binary map. The ground truth binary map is generated based on whether the neighborhood pixel is from the same class as the central pixel. Similar architectural blocks have been assigned the same color scheme for clarity, while the outputs have different colors to differentiate them.

in these methods, generating potent superpixel maps requires the full HSI cube, which can cause data leakage about the test cases. To overcome these limitations, a novel dual-branch model is proposed in this study that avoids the pitfalls of utilizing dimensionality reduction methods in forming the initial graph, while retaining the advantages of incorporating graph-based methods to aid the spatial modeling of 3-D HSI input.

First, a 3-D-CNN with multiscale pyramid attention is proposed to take advantage of the local dependency processing capabilities of CNN (to process spatial features) and the long-range dependency processing capabilities of the attention mechanism (specially for spectral features). A GNN-based network is proposed that will generate an attention map based on the similarity between central and adjacent pixels to overcome spatial neighborhood biases. This attention map is passed to the intermediate layers of the 3-D-CNN to better model the features based on the new information. The attention map generation process itself is supervised using a loss function based on the class label of the central pixel and adjacent pixels, which allows for a more refined attention map that can aid in avoiding neighborhood biases. The proposed architecture consists of the following contributions.

1) A 3-D spectral–spatial feature network (3D-S2FN) is proposed that employs pyramid squeeze attention to selectively prioritize features across spectral, spatial, and spectral–spatial dimensions of the surrounding areas of the desired pixel and generate feature vectors.
2) A graph attention feature processor (GAFP) is proposed that modifies the feature matrices of the 3D-S2FN block using its graph attention map (GRAM). Additionally, this module generates a feature vector by comparing the central pixel and surrounding pixels.
3) An additional loss-based supervision scheme is proposed that seeks to alleviate the issues of overfitting and introduce the neighborhood information in the model.

II. Related Work

Initial methods of HSI classification, such as SVM, PCA-based methods, and local discriminant embedding approaches relied on manual feature engineering [25], [26], [27]. However, manual feature selection requires considerable time and effort, and without proper feature selection, these methods do not perform well. Deep learning-based models eventually replaced them with their superior performance and the elimination of the need for manual supervision in feature selection or generation. In [28], 1-D-CNN layers are utilized for spectral processing and HSI classification. Subsequently, in [7], 2-D-CNN-based models are applied for HSI classification. In [8], a multiscale filter-cascaded contextual CNN is proposed to explore the local spatial relationships in HSI. The potential of 3-D-CNN layers for spectral and spatial processing is demonstrated in [29]. In [10], spectral–spatial residual connection is used along with 3-D-CNNs to extract features for HSI classification.

The self-attention mechanism has been applied to the spectral and spatial domains to classify HSI in multiple studies [14], [30]. This mechanism projects an input X into three linear subspaces, and the projections (Q, K, and V) are utilized to calculate relative feature importance using the equation below, where d_k is the dimension of the keys (or queries).

$$f_{\text{Attention}}(Q, K, V) = f_{\text{softmax}}\left(\frac{QK^{\text{T}}}{\sqrt{d_k}}\right)V. \tag{1}$$

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The methods mentioned rely on modeling the HSI classification as a task where 3-D hyperspectral data cubes are used as input and the central pixels of the cubes are the classification targets. But, as noted in [31], CNN-based models are prone to overoptimistic predictions when randomly sampled train and test sets are used, as there might be overlaps between non-target pixels in train and test sets in randomly sampled train–test sets. This overlap can cause data leakage in the model and cause the model to learn test features, leading to overoptimistic performance on test sets. In contrast, in disjoint train–test sets, there is ample room for performance improvements that a CNN model alone, with its distinct limitations, may not achieve and could thus benefit from incorporating features from alternative model architectures. In [15], it is noted that CNN-based models have difficulty retaining boundary structure information and classifying cases when the central pixel has pixels from differing classes around it, and a method to support CNN models using GNN is proposed.

Unlike CNNs, GNNs perform computations to directly model connections between data. For instance, graph convolutional networks (GCNs) utilize the following equation to update features based on their connections:

$$X_{i+1} = \sigma\left(D^{-\frac{1}{2}}\hat{A}D^{-\frac{1}{2}}X_i W_i\right). \tag{2}$$

Here, X_{i+1} represents the updated features, which are derived from the previous features X_i based on the connections between nodes utilizing adjacency matrix $\hat{A}$, degree matrix D, and learnable weights W. This process, capable of capturing connections between features, has also been applied in HSI classification. In [17], parallel graphs with varying depths are constructed based on HSI, and the classification results are improved by utilizing updated graph information. A semisupervised learning framework with a fixed graph is designed in [32]. In [17], a multiscale graph neural network (GNN) is proposed to process HSI information. In [33], nonlocal GCN is applied to use the whole image as input. These methods are computationally feasible by constructing an initial graph using SLIC-based superpixel generation [21] or CNN-based feature reduction [19]. But, as noted in [34], having too many graph nodes can require high computational complexity, while utilizing superpixel segmentation methods can cause loss of information about the central pixel. To address this issue, a multilayer graph attention network is used in [34] to adaptively learn graph structure for improved classification. In [35] and [36], superpixels are formed first, incorporating mean spectral values of superpixels from the main HSI image as superpixel features, and then graph clusters are formed. To reduce the over-smoothing problem of GCNs, a dense structure along with an auto-regressive moving average (ARMA) filter is proposed in [37] for handling noise better. These methods began with SLIC and then attempted to mitigate the issues that arose out of it.

In [38], the GCN graph is adaptively refined by using a cuckoo search strategy (CSS) [39] to help the network focus more on the significant features. Clustered features are generated in [40] by aggregating local–global correlations and a kernel-based extreme learning machine is employed on the clusters to classify HSI. Aside from GNN-based methods, utilizing information from graphs to aid another network in the classification process has been a strategy for increasing classification performance. A combination of CNN and GNN is utilized in [16] and [18] for improved classification. In contrast, CNN modules are used in [19] to extract initial graph features, which are fine-tuned by subsequent graph convolutions. In [41], a local window approach is employed to process graph structure embeddings, along with an absolute positional embedding mechanism to model global and local features through a transformer network.

III. Methodology

Let $X_{\text{orig}} \in \mathbb{R}^{H\times W\times B}$ be an HSI cube with height H, width W, and number of hyperspectral bands B. The HSI pixels in X_{orig} are grouped into L_c land-cover classes represented by $Y = y_1, y_2, \ldots, y_{L_c}$. The joint spectral and spatial information is used by considering the regions of size $S \times S$ centered at pixel (i, j) (where i and j are positions in height and weight dimensions, respectively), which can be defined as a spectral–spatial matrix of dimensions $S \times S \times B$. Stacking all such matrices for ($i = 1, 2, \ldots, H$ and $j = 1, 2, \ldots, W$), we obtain a tensor $\mathbf{X} \in \mathbb{R}^{N\times S\times S\times B}$, where N is the total number of samples possible. For each spectral–spatial matrix $X_n \in \mathbf{X}$ ($n = 1, 2, \ldots .N$), there is an associated ground truth label $y_{c_n} \in Y$, which is the class label of the central pixel of X_n. Additionally, for each $X_n \in \mathbf{X}$ ($n = 1, 2, \ldots .N$), a binary class mask $y_{\text{mask}_n} \in \mathbf{R^{S\times S}}$ is created based on the ground truth labels y_{ij} for all the pixels in X_n. The binary mask y_{mask_n} is created by passing all the pixels in X_n to the following function:

$$f_{\text{map_generation}}(y_{ij}) = \begin{cases} 1, & y_{ij} = y_{c_n} \\ 0, & y_{ij} \neq y_{c_n}. \end{cases} \tag{3}$$

The generated $y_{\text{mask}_n} \in \mathbf{R^{S\times S}}$ is a representation of the similarity between the central pixel and the adjacent pixels in X_n. By predicting this similarity, the proposed network is trained to recognize when utilizing the neighborhood information would be suitable.

If $\mathbf{X}$ is passed to our proposed network $f(\cdot)$ with parameters θ, the primary objective is to minimize the cost function $L_{\text{obj}}(\theta)$, which is defined as the weighted difference (with α as the parameter for weighting) between the predicted outputs $\hat{y}_{c_n}$ and $\hat{y}_{\text{mask}_n}$, and the ground truth labels y_{c_n} and y_{mask_n} corresponding to all n samples of $X_n \in \mathbf{X}$. The optimization objective is given below:

$$\hat{y}_{c_n}, \hat{y_{\text{mask}_n}} = f(X_n; \theta) \tag{4}$$

$$L_{\text{class}}(\theta) = \frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{L_c}\mathcal{L}\left(\theta, \hat{y}_{c_n}, y_{c_n}\right) \tag{5}$$

$$L_{\text{mask}}(\theta) = \frac{1}{N}\sum_{n=1}^{N}\mathcal{L}_{\text{mask}}\left(\theta, \hat{y_{\text{mask}n}}, y_{\text{mask}_n}\right) \tag{6}$$

$$L_{\text{obj}} = \alpha L_{\text{class}} + (1-\alpha)L_{\text{mask}}. \tag{7}$$

The network is trained using backpropagation to adjust the parameters θ to minimize this cost function. In the experiments conducted, α is set as 0.8 to emphasize the classification task.

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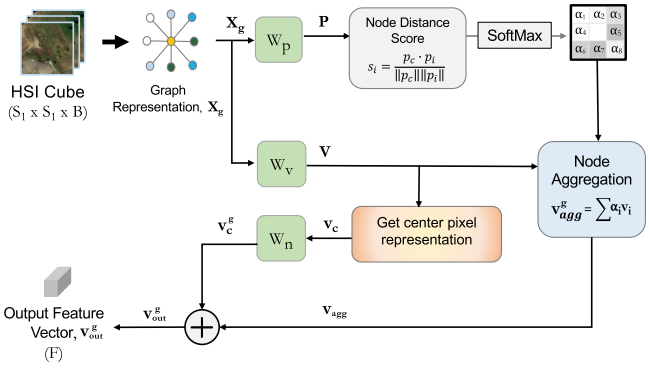


Fig. 2. Proposed GFP submodule. The HSI cube has been represented as a graph, with the central pixel as the central node. The pixels have been projected to linear spaces (using learnable weights W_p and W_v). The similarity between the central pixel and adjacent pixels is utilized to get a node distance score and to aggregate the adjacent features in a feature vector. The central feature vector is additionally processed using a linear layer (denoted by its weights W_n). The adjacent feature vector is added to the processed central feature vector to generate the output feature vector.

The proposed network consists of a 3D-S2FN and a GAFP, which are based on CNNs and GNNs, respectively. Existing methods that solely rely on 3-D-CNNs are limited by the feature extraction capability of 3-D convolutional kernels. For example, these networks can be biased toward the surrounding spatial area of the central pixel for classification, which may reduce their performance in unseen test cases. To overcome this problem, this dual-branch classification network (shown in Fig. 1) is proposed which incorporates 3-D spectral–spatial processing modules and graph-based nodal aggregation modules in parallel branches.

A 3-D HSI volume of spatial size $S \times S$ is provided as input to the proposed 3D-S2FN in Section III-A2. The GAFP is introduced to find the optimum representation of the spatial information for classification. These branches are utilized to generate feature vectors based on their distinct processing capabilities. Finally, these feature vectors are concatenated and processed by a fully connected block to generate the final classification output. Apart from this conventional loss optimization-based supervised classification scheme, an additional loss-based supervision scheme is proposed, namely the binary class map prediction scheme.

In the proposed binary class map prediction scheme (Section III-B), the model is trained to predict if the central HSI pixel and the adjacent pixels are from the same class or not. The attention maps from the proposed GAFP module are used to generate a class map. This class map is trained against a class map generated from HSI cube ground truth labels.

A. Conventional Supervised Learning Scheme

1) Proposed GAFP: Graph-based feature representation methods model connected feature vectors as nodes and links, and use this information to generate potent feature vectors through aggregation. Recent works [38], [42] can estimate the optimum connections between nodes for improved classification performance. The information attained from constructing the relation between the central and adjacent pixels as links, and finding their optimum distances can be highly relevant in HSI classification and the GAFP module is proposed to incorporate this information in the model. Aside from directly contributing a feature vector to aid in classification (denoted as v^g in Fig. 1), this module also introduces a graph attention mask (GRAM) into the 3D-S2FN module to mitigate the neighborhood bias in CNN models.

As shown in Fig. 1, the GAFP module consists of two parallel graph feature processor (GFP) submodules. One of the submodules is used to aggregate the immediately adjacent information in the $S_1 \times S_1$ HSI cube ($S_1 < S$) at the center of the input (formed by taking a local window size of $S_1 \times S_1$), and another is used to aggregate the information from the entire $S \times S$ input HSI cube (local window size $S \times S$). This method facilitates the integration of information across multiple fields of view. By adding more weight to the feature vector generated from immediately adjacent HSI cube, closer features are prioritized.

The GFP submodule (in Fig. 2) models the input HSI cube as a graph where the nodes represent HSI pixels with B number of bands. Similar to the constraint scheme in [43], a local window is formed based on the input HSI cube as input. Initially, all the nodes adjacent to the central HSI node are initialized as equidistance from it, and the submodule is trained to generate node feature vectors and update the nodal distances (the links) between them so that this information can be aggregated for improved performance.

The GFP submodule takes in an input $X_g \in \mathbf{R}^{\mathbf{S_1} \times \mathbf{S_1} \times \mathbf{B}}$ and projects it in shared projection spaces utilizing linear layers (denoted by W_p and W_v, which are the learnable weights associated with the linear layers), with the resultant outputs V and P having the shape $S_1 \times S_1 \times F$, where F is the feature dimension. The central HSI vector representations (shaped F) are extracted from these projections (the operation is denoted by $f_{\text{center_pixel}}$) as follows:

$$P = W_p \cdot X_g, \quad p_c = f_{\text{center_pixel}}(P) \tag{8}$$

$$V = W_v \cdot X_g, \quad v_c = f_{\text{center_pixel}}(V). \tag{9}$$

From all the pixel features $p_i \in P$, the central pixel feature p_c is compared with adjacent HSI vectors p_i (where $i \neq c$) utilizing cosine similarity score to generate an attention map of shape $S_1 \times S_1$, which can be interpreted as the node distance score from central HSI node. This node distance score is an estimation of the similarity between the central pixel feature vector and adjacent pixel feature vectors. This information can be utilized to weight the importance of adjacent pixels in classifying the central pixel class as follows:

$$s_i = \frac{p_c \cdot p_i}{\|p_c\| \|p_i\|}, \quad \text{for } i \neq \text{center} \tag{10}$$

$$\alpha_i = \frac{e^{s_i}}{\sum_j e^{s_j}}. \tag{11}$$

Here, the index of the central HSI vector is denoted using center. The cosine similarity score, s_i, is processed through a softmax operation to get attention scores, α_i. These attention scores or the node distance scores help construct an updated graph, where the distance from the central node feature vector from other feature vectors represents their feature similarity.

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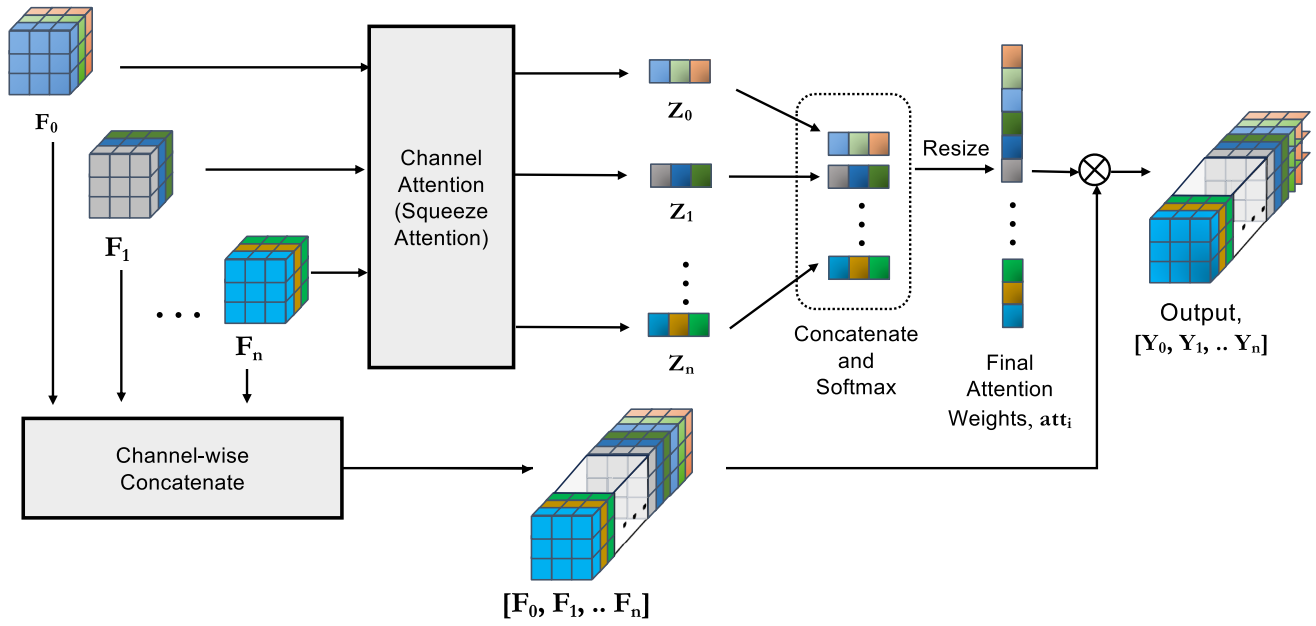


Fig. 3. EPSA block. The number of inputs in the block depends on which module the EPSA submodule is being used. For the proposed SSPSA module, it takes three inputs (which is shown here), and for the proposed residual EPSA, it takes four inputs.

This feature similarity is useful in the context of HSI classification, as it contains fine-grained information about the similarity between the central and adjacent pixels.

The vectors $v_i \in V$ in (9) are updated using the attention scores in (11), generating the feature vector v^g_{agg} where the values are scaled based on their similarity to the central HSI pixel.

$$v^g_{\text{agg}} = \sum_i \alpha_i v_i, \qquad \text{for all } i \neq \text{center}. \tag{12}$$

Since the central node feature vector is more significant in the classification task, a linear layer, denoted by W_n, is dedicated to processing its features more extensively to generate v^g_c of shape F.

$$v^g_c = W_n \cdot v_c. \tag{13}$$

Experimental results indicate that not utilizing this linear layer deteriorates performance, which can be attributed to the importance of additional processing of the central HSI band in the classification process.

The adjacent feature vector, v^g_{agg} [in (12)], is added to the processed central feature vector, v^g_c, to generate the output to the GFP submodule, the output feature vector v^g_{out}.

$$v^g_{\text{out}} = v^g_c + v^g_{\text{agg}}. \tag{14}$$

In summary, the GFP submodule takes in X_g as input and outputs a feature vector v^g_{out}. The intermediate attention scores α_i are also used as a binary mask during training. The GAFP module consists of two parallel GFP submodules that process the input in two scales. The binary mask generated from the $S \times S$ input HSI cube is used as the GRAM in the proposed binary map prediction method (detailed in Section III-B), enabling explicit modeling of the adjacent HSI pixel information to estimate the distance score.

2) Proposed 3-D Spectral–Spatial Feature Network: Combining spatial and spectral information is a widely used approach for producing effective feature vectors in the context of HSI classification. While existing attention-based CNN models can process local spatial information through CNN kernels and long-range dependencies in spectral information using attention mechanism, they can still be affected by spatial neighborhood bias. Therefore, a 3-D Spectral-Spatial Feature Network (3D-S2FN) is proposed that can incorporate information the proposed GAFP module to mitigate this issue. The 3D-S2FN module uses efficient pyramid squeeze attention (EPSA) blocks (shown in Fig. 3) in its two submodules: the spatial–spectral pyramid squeeze attention (SSPSA) submodule and the residual efficient pyramid squeeze attention (residual EPSA) submodule, depicted in Figs. 4 and 5, respectively. The 3D-S2FN module processes the input HSI cube (shaped $S \times S \times B$) by passing it through the SSPSA module and a cascade of residual EPSA modules to generate a feature vector v_s of shape F (as shown in Fig. 1) as output.

The proposed SSPSA submodule (Fig. 5) takes the initial HSI cube (denoted by X in (15) and Fig. 5) as input. $X \in \mathbf{R}^{\mathbf{S}\times\mathbf{S}\times\mathbf{B}}$ is first expanded into multiple channels by applying a 3-D convolutional kernel (convolutions are denoted by $f^i{}_{\text{Conv3D}}$, where i indexes a specific 3-D convolutional layer in the network architecture, whereas parallel convolutions are denoted using $f^{i,j}_{\text{Conv3D}}$, with j denoting the parallel indices), processes it in both the spatial and spectral domains.

$$H = f^1{}_{\text{Conv3D}}(X). \tag{15}$$

The resulting output, H, is then split channel-wise into three smaller matrices (H_1, H_2, and H_3, each having same spatial dimension, but 1/3 of channels) that are fed into three parallel convolutional kernels [in (17)] that each prioritize different features: spatial, spectral, or spatial–spectral. The outputs

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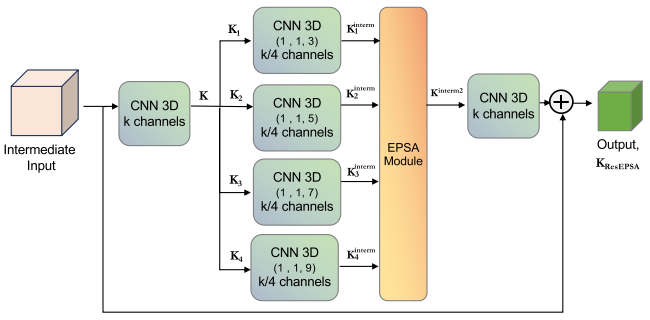


Fig. 4. Residual EPSA module. The input is passed through a CNN layer with k channels, and the resultant output channels are split by four and passed to CNN layers with varying spectral kernel lengths (3, 5, 7, and 9). The EPSA submodule weights the relative importance of the processing done by each CNN layer before aggregating them. A residual connection adds the input to the output.

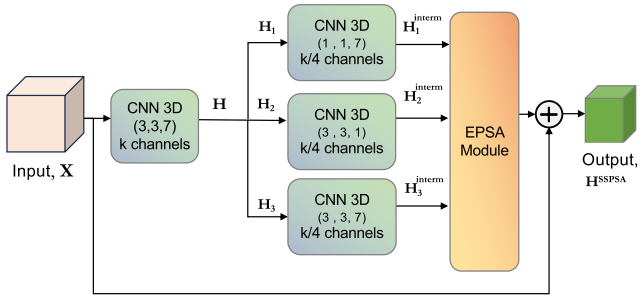


Fig. 5. Proposed SSPSA module. The initial input is processed spectrally and spatially before passing on to three CNN filters with differing kernel shapes. Their outputs are weighted using the EPSA module. A residual connection is also added to the resultant output.

of these kernels are subsequently passed through the EPSA block [44] (denoted by $f^{i}{}_{\text{EPSA}}$, with i indexing a specific EPSA block), shown in Fig. 3, which assigns attention weights to features. This enables the model to learn which feature (spatial, spectral, or both) to emphasize for a given input.

$$H = [H_1, H_2, H_3] \quad \text{(Channelwise split)} \tag{16}$$

$$H_j^{\text{interm}} = f_{\text{Conv3D}}^{2,j}\left(H_j\right) \tag{17}$$

$$H^{\text{SSPSA}} = f^{0}{}_{\text{EPSA}}\left(\left[H_1^{\text{interm}}, \ldots H_3^{\text{interm}}\right]\right) + X. \tag{18}$$

The residual EPSA submodule (Fig. 4) extends the EPSANet [44] to 3-D blocks by processing its input across spectral kernels on multiple scales. Given the output from the SSPSA submodule, H^{SSPSA} is given as follows:

$$K = f^{3}{}_{\text{Conv3D}}\left(H^{\text{SSPSA}}\right) \tag{19}$$

$$K = [K_1, K_2, K_3, K_4] \quad \text{(Channelwise split)} \tag{20}$$

$$K_j^{\text{interm}} = f_{\text{Conv3D}}^{4,j}\left(K_j\right) \tag{21}$$

$$K^{\text{interm2}} = f^{1}{}_{\text{EPSA}}\left(\left[K_1^{\text{interm}}, \ldots, K_4^{\text{interm}}\right]\right) \tag{22}$$

$$K^{\text{ResEPSA}} = f^{5}{}_{\text{Conv3D}}\left(K^{\text{interm2}}\right) + H^{\text{SSPSA}}. \tag{23}$$

The resultant output K^{ResEPSA} is passed to subsequent ResEPSA layers as inputs. The output of the second ResEPSA layer [denoted as $X^{s}{}_{\text{masked}}$ in (27)] is multiplied with GRAM before being passed to the final ResEPSA layer.

The EPSA block used in both SSPSA and ResEPSA (Fig. 3) takes multiple inputs and utilizes pyramid attention mechanism to weight their channels. $[H_1^{\text{interm}}, \ldots H_3^{\text{interm}}]$ [on (18)] and $[K_1^{\text{interm}}, \ldots, K_4^{\text{interm}}]$ [on (22)] are used as inputs to this block in SSPSA and ResEPSA, respectively. These inputs are generalized as $F_1, F_2, \ldots, F_n$ in the following equation, and f_{ReLU} is denoted as the ReLU function:

$$Z_i = \sigma\left(W_2 \cdot f_{\text{ReLU}}(W_1 \cdot F_i)\right) \cdot F_i \tag{24}$$

$$\text{att}_i = \frac{e^{Z_i}}{\sum_j e^{Z_j}}, \quad Y_i = F_i \odot \text{att}_i \tag{25}$$

$$\text{out} = [Y_1, \ldots, Y_n]. \tag{26}$$

The proposed 3D-S2FN module comprises an initial SSPSA block and a cascade of three residual EPSA submodules. The CNN layers at the start and end of the ResEPSA submodules have varying kernel sizes that are tailored to focus on a specific dimension. For example, the first residual EPSA submodule has convolutional kernels with a shape of (1, 1, 7), which allows it to focus on the spectral dimension. Subsequent residual EPSA submodules use kernels that prioritize spatial (3, 3, 1) and spectral–spatial (3, 3, 7) features, respectively.

The GRAM produced by the proposed GAFP is multiplied by the output of the spatial residual EPSA submodule (denoted as intermediate output, $X^{s}{}_{\text{interm}}$):

$$X^{s}{}_{\text{masked}} = X^{s}{}_{\text{interm}} \odot \text{GRAM}. \tag{27}$$

This operation modifies the output features using neighborhood information processed by GAFP. As this output is eventually used to generate predictions and train the network through backpropagation, this collaborative process enables the two modules to work together to generate effective features.

The fully connected block takes the concatenated feature vectors from GAFP and 3D-S2FN and passes them through a series of fully connected layers to model the information available from them and finally generate a classification prediction.

B. Proposed Binary Class Map Prediction Scheme

Although the main objective is to predict the central HSI pixel, the information regarding the surrounding areas of the pixel can still be useful. Especially, whether the central HSI pixel and the adjacent HSI pixels are of the same class or not is a highly relevant information for the improved classification of the central HSI pixel. In the classification process, if the surrounding pixels are from different classes, it is conceivable that they may potentially misguide the classifier, resulting in an erroneous classification of the central pixel. In response to this concern, the node distance map generated by the GAFP module, which is based on the similarity between the central pixel and its surrounding pixels, is utilized as a binary segmentation map. This map is passed to the 3D-S2FN to function as an attention map. This attention map instructs the network regarding the degree of importance that should be accorded to the surrounding pixels in the classification process. Consequently, the binary segmentation map serves the purpose

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Algorithm 1 Pseudocode of the Proposed SSGRAM Method

Input: Training samples X (shape $S \times S \times B$), Training class labels Y
Output: Test Labels
/* Training Procedure */
for epoch $\in \{1, \ldots,$ Epochs$\}$ **do**
  for each batch (X, Y) **do**
    /* GAFP Feature */
    $X_{\text{center}} \leftarrow$ slice a smaller cube of size $S_1 \times S_1 \times B$ from the center of X $(S_1 < S)$
    $v_c^g \leftarrow$ pass X_{center} through GFP via (8)-(14)
    v_{all}^g, GRAM $\leftarrow$ pass X through GFP via (8)-(14)
    $v^g \leftarrow v_{\text{all}}^g + v_c^g$
    /* 3D-S2FN Feature */
    $X_{\text{interm}}^s \leftarrow$ pass X through SSPSA and 2 Residual EPSA layers via (15) - (23)
    $X_{\text{masked}}^s \leftarrow X_{\text{interm}}^s \odot$ GRAM
    $X_{\text{final}}^s \leftarrow$ pass X_{masked}^s through 1 Residual EPSA layer via (19) - (23)
    $v^s \leftarrow$ Pooling operation on X_{final}^s
    /* Combining Features */
    $Y_{\text{pred}} \leftarrow$ pass $[v^s, v^g]$ to fully connected layer
    /* Loss Calculation */
    $l_{\text{class}} \leftarrow$ use Y_{pred} and Y to calculate class loss
    $Y_{\text{mask}} \leftarrow$ use Y to build binary map using (28)
    $l_{\text{map}} \leftarrow$ use GRAM and Y to calculate mask loss
    $l \leftarrow l_{\text{class}} + l_{\text{map}}$
    minimize l and update parameters
  end for
end for
/* Testing Procedure */
Predict Test Labels utilizing the trained model

of mitigating the influence of surrounding pixels when they are different from the target classification.

In this approach, the binary segmentation map is trained against a target binary map that is constructed from the HSI cube class labels. To construct this target binary map, the class labels for the entire input HSI cube are utilized. The method of generating the target binary map out of the class labels is as follows:

$$f_{\text{map_generation}}(y_{ij}) = \begin{cases} 1, & y_{ij} = y_c \\ 0, & y_{ij} \neq y_c \end{cases} \tag{28}$$

where y_{ij} is the class label of the HSI pixel located at ith row and jth column of the input HSI cube, and y_c is the class label of the central HSI pixel. This generated binary map is used to additionally supervise the binary segmentation map or the attention map generated by the proposed GAFP module. The attention map is also applied on the feature map of the intermediate submodules in the proposed 3D-S2FN module [described in detail in Section III-A2]. The binary class map prediction scheme explicitly models the information available from the surrounding HSI pixels. The complete pseudocode of the proposed GRAM method is provided in Algorithm 1.

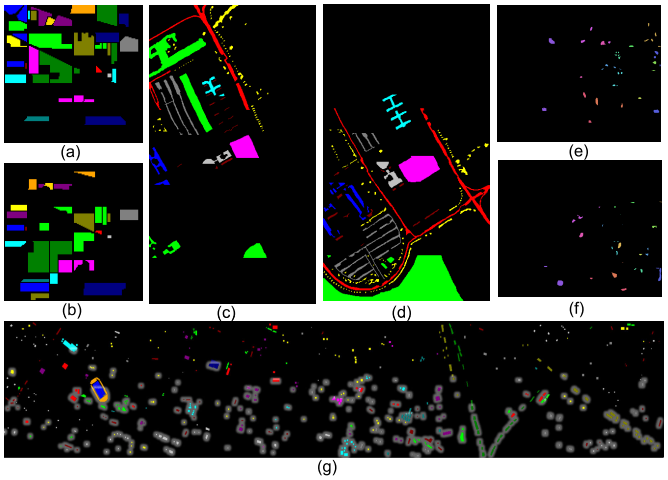


Fig. 6. Training and testing set representations. (a) and (b) Disjoint IP dataset. (c) and (d) Disjoint PU dataset. (e) and (f) Disjoint KSC dataset. (g) Disjoint Houston 2013 dataset. For the Houston dataset, the testing set data points are shown with a white glow around its border for ease of visualization.

IV. Experimental Results

A. Datasets

The proposed solution is trained and evaluated on the widely used Indian Pines (IP), Kennedy Space Center (KSC), Pavia University (PU), and Houston 2013 (HU) HSI datasets. To avoid over-optimistic results caused by overfitting, disjoint sets of the datasets are used to train and evaluate the performance. Fig. 6 provides the ground truth details for the datasets. The disjoint sets are generated using the DeepHyperX toolbox introduced in [31]. The performance is also compared on randomly sampled train and test sets consisting of 10% and 90% of the datasets, respectively.

1) Indian Pines: The IP dataset consists of 145 × 145 pixels and 200 spectral reflectance bands gathered from AVIRIS [45] sensor over the IP test site in Northwest Indiana. This dataset consists of 16 classes.

2) Pavia University: The PU dataset is acquired by ROSIS [46] sensor during a flight campaign over Pavia, Italy. This dataset was originally a 610 × 610 pixels image consisting of 103 spectral bands, but some samples in the images contained no information and had to be discarded. After discarding, the image resolution is 610 × 340 pixels, which is used as the dataset. There are nine classes in the dataset.

3) Kennedy Space Center: The KSC dataset is acquired by AVIRIS [45] sensor. This dataset has 512 × 614 pixels with each pixel containing 176 spectral bands (after removing noisy bands). There are 13 classes in the dataset.

4) Houston 2013: The Houston 2013 Dataset consists of 349 × 1905 pixels and 144 spectral bands acquired over the University of Houston campus and the neighboring urban area. The dataset contains 15 classes.

B. Experimental Setup

All the networks are implemented on the PyTorch platform. The equipment used in the following experiments consists of

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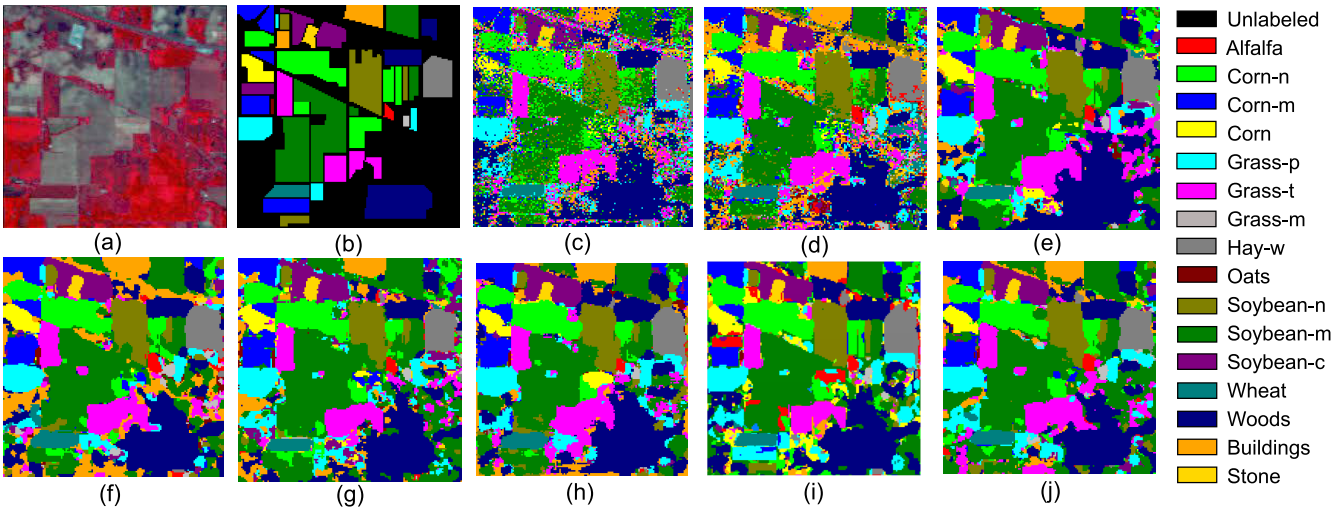


Fig. 7. Classification map for disjoint IP dataset. (a) False-color composite image. (b) Ground truth. (c)–(j) Classification maps produced by 1-D-CNN, 2-D-CNN, SSRN, A^2S^2k-ResNet, S^{2}RGANet, RIAN, RepSSRN, and the proposed method, respectively.

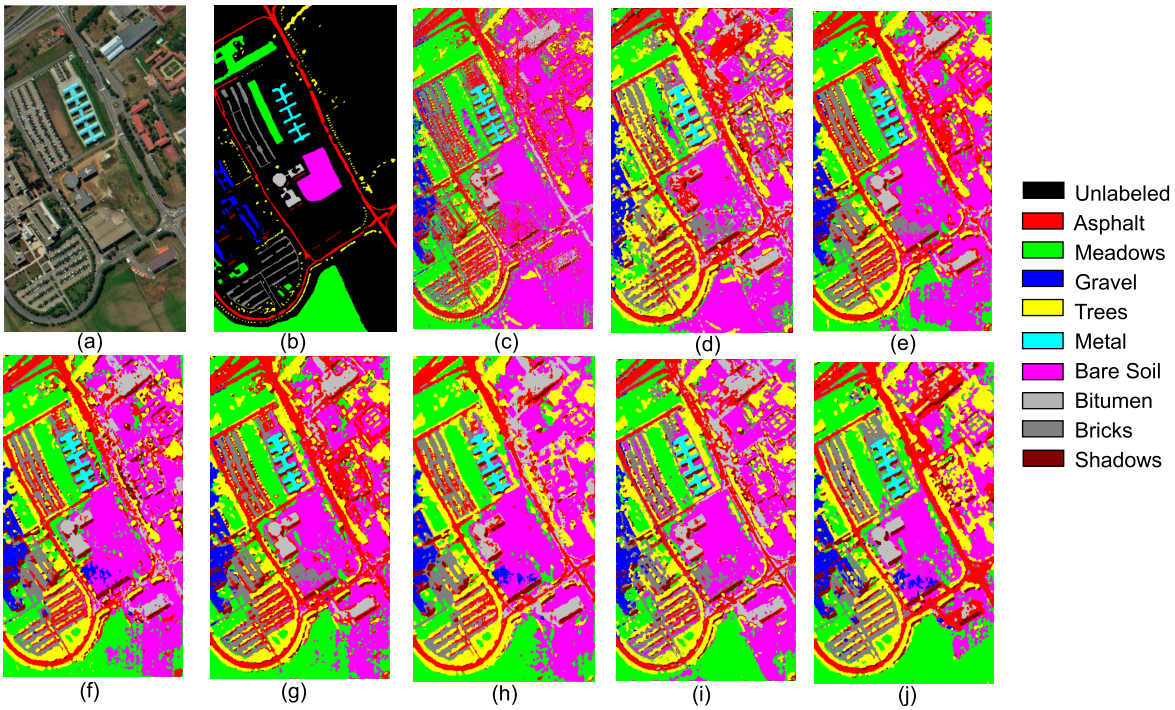


Fig. 8. Classification map for disjoint PU dataset. (a) False-color composite image. (b) Ground truth. (c)–(j) Classification maps produced by 1-D-CNN, 2-D-CNN, SSRN, A^2S^2k-ResNet, S^{2}RGANet, RIAN, RepSSRN, and the proposed method, respectively.

an Intel[1] Xeon[1] CPU @ 2.00 GHz and one NVIDIA Tesla P100 with 16 GB of memory. The hyperspectral cubes with a size of $7 \times 7 \times C$ are used as the input. The class of the input sample is determined by the center pixel of the cube. The proposed method is trained for 50 epochs using the Adam optimizer with a learning rate $\alpha = 0.001$. For the other baselines, the parameters reported in their respective papers are adopted. The classification results are evaluated quantitatively with four metrics: class-wise accuracy, average accuracy (AA), overall accuracy (OA), and the kappa coefficient (KC).

[1]Registered trademarked.

To demonstrate the effectiveness of the proposed method, the performance of existing state-of-the-art (SOTA) methods with various architectures is compared: method that only use central pixel (1-D-CNN [28]), 2-D-CNN architecture-based method (2-D-CNN [47]), Spectral–Spatial Residual Network (SSRN [10]) and reparameterized RepSSRN [48], Graph attention-based S^{2}RGANet [18], Attention-based A^2S^2k-Resnet [49] and RIAN [30] are picked for comparison.

C. Results and Discussion

1) Disjoint Datasets: The proposed method outperforms existing methods on disjoint sets of four datasets, with

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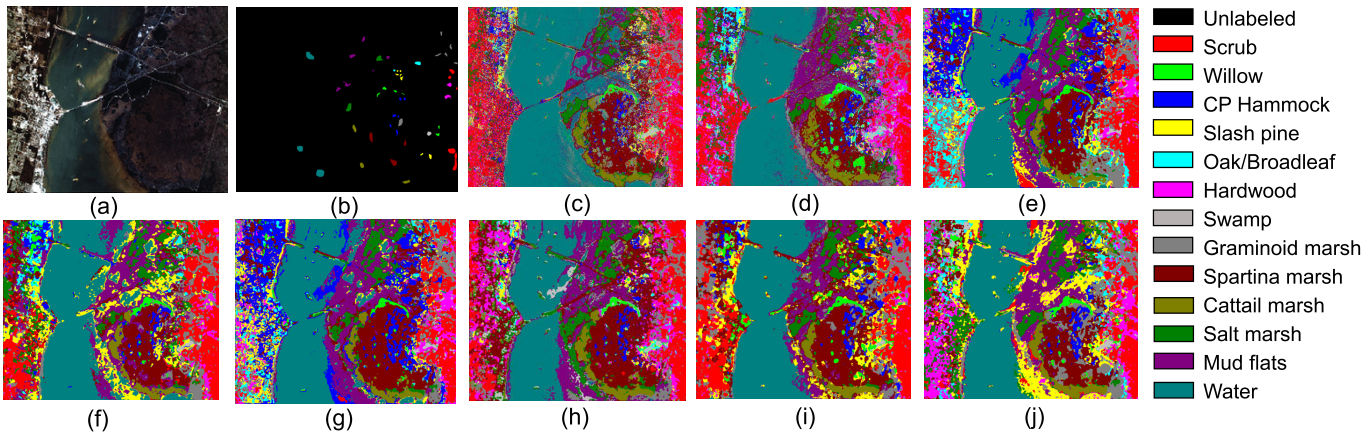


Fig. 9. Classification map for disjoint KSC dataset. (a) False-color composite image. (b) Ground truth. (c)–(j) Classification maps produced by 1-D-CNN, 2-D-CNN, SSRN, A^2S^2k-ResNet, S^{2}RGANet, RIAN, RepSSRN, and the proposed method, respectively.

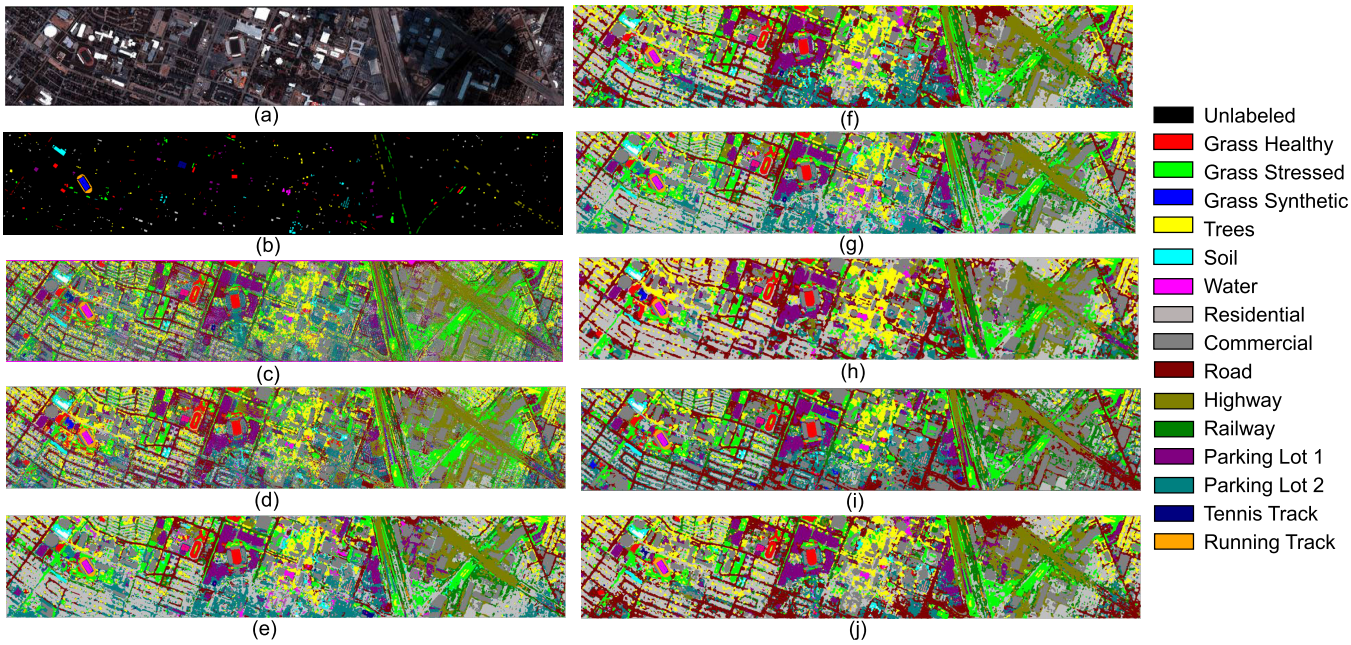


Fig. 10. Classification map for disjoint Houston 2013 dataset. (a) False-color composite image. (b) Ground truth. (c)–(j) Classification maps produced by 1-D-CNN, 2-D-CNN, SSRN, A^2S^2k-ResNet, S^{2}RGANet, RIAN, RepSSRN, and the proposed method, respectively.

particularly noteworthy relative performance in the PU and Houston 2013 datasets.

a) IP dataset: Experiments on the IP dataset (containing 16 classes with varying sample sizes) are shown in Table I. It can be observed from Fig. 7 that the models can accurately predict areas where the surrounding areas have the same class, but in smaller areas where different classes are together, the models face difficulty accurately predicting the class. The SSGRAM method performs comparatively better in these areas, which can be attributed to the usage of GRAM. While existing SOTA methods sometimes outperform the proposed method on a few classes, they still cannot maintain performance across all classes, whereas the proposed method achieves better results overall.

b) PU dataset: Table II shows the experiments on the PU dataset comparing existing and proposed approaches. In Fig. 8, it can be observed that other models tend to confuse meadows with bare soil due to their similar characteristics, where the proposed method performs better. Despite having some difficulty differentiating bricks and bitumen, it still outperforms existing models.

c) KSC dataset: The reported results in Table III showcase the comparison of existing and proposed approaches for experiments conducted on the KSC dataset. The table and Fig. 9 demonstrate that the models tend to confuse graminoid and Spartina marsh and have relatively lower class accuracy for them. Although oversampling and augmentation techniques are used to address this issue, they do not entirely alleviate the problem. It is apparent from the table that the proposed method outperforms existing approaches in this dataset as well.

d) Houston 2013 dataset: In Table IV, the result comparison between the proposed method and other existing methods is shown. Houston 2013 is the largest among all the datasets

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TABLE I
COMPARISON OF PERFORMANCES ON DISJOINT IP SCENE

Class	Methods							
	1-D CNN	2-D CNN	SSRN	A^2S^2k-ResNet	S^2-RGANet	RIAN	RepSSRN	Proposed
1	0.726	0.668	0.913	0.9	0.872	*0.921*	0.9	**0.93**
2	0.783	0.693	0.812	0.813	**0.831**	0.791	0.673	*0.824*
3	0.62	0.572	*0.791*	0.721	0.401	0.784	**0.801**	0.726
4	0.334	0.261	0.818	0.874	0.852	**0.91**	0.894	*0.907*
5	0.847	0.626	0.873	0.822	0.862	*0.905*	0.881	**0.912**
6	0.937	*0.978*	**0.99**	0.854	0.891	0.91	**0.99**	0.894
7	0.57	0.3	0.642	0.981	0.867	*0.985*	0.912	**0.992**
8	0.941	0.931	0.96	0.95	0.881	*0.976*	0.971	**0.987**
9	0.535	0.756	0.771	0.677	0.802	**0.837**	0.813	*0.821*
10	0.742	0.806	0.831	0.87	**0.93**	0.894	*0.92*	0.852
11	0.79	*0.813*	0.73	0.761	0.713	*0.813*	**0.821**	0.792
12	0.664	0.531	0.9	*0.94*	0.768	**0.961**	0.921	0.914
13	0.961	**0.99**	0.974	0.833	0.901	**0.99**	0.979	*0.983*
14	0.916	0.94	0.972	0.971	0.893	0.973	*0.982*	**0.987**
15	0.823	0.877	*0.989*	0.93	0.403	0.678	**0.99**	0.965
16	0.886	0.902	0.86	**0.998**	0.978	0.952	0.891	*0.983*
OA	0.793	0.775	0.823	*0.848*	0.805	0.837	0.834	**0.854**
AA	0.756	0.730	0.861	0.86	0.80	0.894	*0.8965*	**0.901**
KC	0.764	0.743	0.82	0.812	0.722	*0.827*	0.825	**0.836**

TABLE II
COMPARISON OF PERFORMANCES ON DISJOINT PU SCENE

Class	Methods							
	1-D CNN	2-D CNN	SSRN	A^2S^2k-ResNet	S^2RGANet	RIAN	RepSSRN	Proposed
1	0.424	0.748	0.965	0.923	0.951	0.965	*0.972*	**0.988**
2	0.524	0.568	0.664	0.888	*0.973*	0.891	0.972	**0.994**
3	0.279	0.418	0.893	0.643	0.749	0.894	*0.903*	**0.967**
4	0.829	0.704	0.818	0.814	0.8586	*0.923*	0.844	**0.986**
5	0.774	0.939	**0.998**	*0.985*	**0.998**	0.984	0.978	0.978
6	0.725	0.771	0.921	0.816	0.915	*0.975*	**0.984**	0.973
7	0.786	*0.891*	0.735	0.785	**0.944**	0.794	0.773	0.878
8	0.724	0.85	0.749	**0.878**	*0.864*	0.845	0.831	0.8
9	0.494	0.647	0.89	0.972	*0.987*	0.923	0.915	**0.989**
OA	0.628	0.671	0.875	0.903	0.937	0.912	*0.941*	**0.95**
AA	0.617	0.726	0.848	0.856	*0.915*	0.910	0.908	**0.951**
KC	0.643	0.691	0.831	0.87	0.916	0.912	*0.931*	**0.949**

TABLE III
COMPARISON OF PERFORMANCES ON DISJOINT KSC DATASET

Class	Method							
	1D-CNN	2D-CNN	SSRN	A^2S^2k-Net	S^2RGA-Net	RIAN	RepSSRN	Proposed
1	0.872	0.956	**0.994**	0.982	0.92	0.912	0.9	*0.99*
2	*0.947*	0.65	**1.00**	**1.00**	0.8	0.8	0.857	**1.00**
3	0.774	0.938	*0.973*	0.76	0.933	0.786	**0.987**	0.801
4	0.455	0.512	0.507	*0.579*	0.551	0.554	0.574	**0.629**
5	0.604	0.972	0.895	**1.00**	0.55	*0.985*	0.904	0.84
6	0.536	0.646	0.557	0.574	0.544	0.601	*0.67*	**0.675**
7	0.27	0.298	0.341	0.34	0.297	**0.673**	*0.453*	0.34
8	0.633	0.542	0.541	0.619	*0.89*	0.865	0.764	**0.982**
9	0.531	0.528	0.569	0.62	0.621	**0.653**	0.612	*0.64*
10	0.988	*0.994*	**1.00**	**1.00**	**1.00**	**1.00**	0.99	**1.00**
11	0.973	0.784	**1.00**	**1.00**	0.836	0.785	*0.99*	**1.00**
12	*0.963*	0.959	**1.00**	**1.00**	**1.00**	0.712	**1.00**	**1.00**
13	0.992	**1.00**	**1.00**	**1.00**	*0.997*	**1.00**	0.99	**1.00**
OA	0.781	0.807	0.832	*0.843*	0.822	0.829	0.838	**0.846**
AA	0.734	0.7615	0.798	0.808	0.765	0.794	*0.822*	**0.837**
KC	0.756	0.785	0.814	0.825	0.802	0.821	*0.827*	**0.831**

used to compare the performances. The proposed method surpasses other existing methods in the reported metrics. However, the overall results on this dataset are considerably lower compared to other datasets, which can be attributed to the relatively higher difficulty of this dataset. In Fig. 10, the complete classification map is shown. From this figure and Table IV, it can be observed that the proposed model excels in classifying commercial areas and parking lot type 1 over other models.

Although the proposed method faces some difficulty in classifying roads and parking lot type 2, the overall OA, AA, and KC scores surpass all other methods.

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TABLE IV
COMPARISON OF PERFORMANCES ON DISJOINT HOUSTON 2013 SCENE

Class	Methods							
	1-D CNN	2-D CNN	SSRN	A^2S^2-ResNet	S^2RGANet	RIAN	RepSSRN	Proposed
1	0.925	0.949	0.986	0.983	**1.00**	**1.00**	*0.986*	**1.00**
2	0.724	0.721	*0.828*	0.826	0.822	0.684	**0.845**	0.732
3	*0.998*	0.968	*0.998*	*0.998*	**1.00**	0.993	0.877	0.995
4	0.845	0.718	0.929	**0.998**	0.934	0.921	*0.984*	0.942
5	*0.986*	**0.998**	0.78	0.963	0.781	0.969	0.965	**0.998**
6	0.92	0.891	0.837	**0.978**	0.856	0.911	0.941	*0.961*
7	*0.653*	0.652	0.413	0.503	0.415	0.621	**0.709**	0.592
8	0.408	0.463	*0.674*	0.551	0.656	0.576	0.491	**0.693**
9	0.495	0.576	*0.795*	0.436	**0.817**	0.648	0.414	0.489
10	0.64	0.583	0.88	0.797	*0.892*	0.782	**0.96**	0.83
11	0.499	**0.771**	0.356	0.734	0.358	0.412	*0.77*	0.748
12	0.52	0.453	*0.863*	0.808	0.862	0.48	0.771	**0.941**
13	0.274	0.268	0.304	0.356	0.31	*0.528*	**0.559**	0.416
14	0.851	*0.96*	0.807	0.93	0.823	**1.00**	**1.00**	**1.00**
15	0.986	0.984	0.993	**1.00**	**1.00**	0.956	*0.998*	**1.00**
OA	0.633	0.645	0.654	0.692	0.653	0.7	*0.703*	**0.729**
AA	0.715	0.73	0.763	0.791	0.768	0.765	*0.818*	**0.822**
KC	0.604	0.618	0.628	0.668	0.626	0.676	*0.679*	**0.708**

TABLE V
PERFORMANCE IN RANDOM SAMPLING

Metrics	Methodology							
	1D-CNN	2D-CNN	SSRN	A^2S^2k-Net	S^2RGA-Net	RIAN	RepSSRN	Proposed
Indian Pines (10% Training Samples)								
OA	0.564	0.796	0.978	*0.986*	0.932	**0.987**	0.985	0.981
AA	0.538	0.602	0.858	0.927	*0.962*	0.958	0.913	**0.972**
KC	0.495	0.768	0.974	0.958	0.923	*0.981*	0.979	**0.984**
Pavia University (10% Training Samples)								
OA	0.841	0.921	*0.989*	0.977	0.942	0.985	**0.992**	0.986
AA	0.81	0.912	*0.984*	0.968	0.919	0.98	**0.988**	0.979
KC	0.789	0.916	0.985	0.973	0.91	*0.987*	**0.99**	0.981
Kennedy Space Center (10% Training Samples)								
OA	0.684	0.841	0.981	**0.984**	0.939	0.981	*0.982*	0.98
AA	0.626	0.833	0.896	0.942	0.947	0.949	*0.957*	**0.965**
KC	0.568	0.764	0.978	0.964	0.93	*0.975*	**0.983**	0.95
Houston 2013 (10% Training Samples)								
OA	0.711	0.846	0.988	**0.99**	0.981	0.978	0.988	*0.989*
AA	0.753	0.865	0.986	*0.99*	0.98	0.981	0.986	**0.991**
KC	0.688	0.834	*0.987*	*0.987*	0.979	0.976	*0.987*	**0.988**

TABLE VI
ABLATION STUDY ON DISJOINT PU SCENE

Module Ablation				
Network	Optimization	OA	AA	KC
3D-S2FN	Classification Loss	0.939	0.932	0.943
3D-S2FN+GAFP	Classification Loss	0.944	0.943	0.944
3D-S2FN+GAFP	Classification Loss + Map Loss	**0.95**	**0.951**	**0.949**
3D-S2FN Depth Ablation				
Method	3D-S2FN Layers	OA	AA	KC
3D-S2FN + GAFP (with both losses)	2	0.946	0.9457	0.944
	3	0.95	**0.951**	**0.949**
	4	**0.9509**	0.9503	0.947

2) Random Sampling Method: In this study, model performance is evaluated using both disjoint and randomly sampled train and test sets. Disjoint sets of IP, KSC, PU, and Houston 2013 datasets are utilized for training and evaluation in Tables I–IV. In addition, experiments are also performed using a randomly sampled train and test set, as described in Table V. In this scenario, 10% of the total dataset is utilized for training, while the remaining data is utilized for evaluation.

Notably, the performance on randomly sampled datasets is considerably better compared to their disjoint counterparts, which can be attributed to the over-optimistic results in the random sampling method, a phenomenon extensively studied in [31]. The performance across models is very similar in this method, and there are few noteworthy patterns. The proposed model provides improved performance in the IP and Houston 2013 dataset across AA and KC metrics, but it is notable that, on the PU dataset, RepSSRN [48] performs better across OA, AA, and KC metrics. This increase in performance can be attributed to the relatively higher number of parameters in RepSSRN, enabled by reparameterization) which becomes particularly useful when there are ample training samples, as is the case for the PU dataset.

However, RepSSRN does not perform the best in the largest dataset, Houston 2013, or the smallest dataset, IP. For IP, there are fewer training samples, which may cause overfitting in a high-parameter model and reduce performance. As for Houston 2013, the data points are more spread out and there

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TABLE VII
ABLATION STUDY ON VARIATIONS IN CENTRAL AND INPUT HSI CUBE SIZE ON DISJOINT PU DATASET

Input HSI Cube Size	Central HSI Cube Size											
	3			5			7			9		
	OA	KC	AA	OA	KC	AA	OA	KC	AA	OA	KC	AA
7	0.95	0.949	**0.951**	0.951	0.9541	0.9448	X			X		
9	0.9473	0.9495	0.9494	**0.9526**	**0.9757**	0.9336	0.9498	0.953	0.9446	X		
11	0.947	0.948	0.942	0.9475	0.95	0.943	0.947	0.9493	0.9434	0.9494	0.9525	0.9294

TABLE VIII
ABLATION STUDY ON THE EFFECT OF 3D-S2FN AND GAFP ON DISJOINT KSC DATASET

Class	GAFP	3D-S2FN	Proposed (3D-S2FN + GAFP)
1	0.9641	0.9873	**0.99**
2	0.9759	0.9952	**1.00**
3	**0.804**	0.7948	0.801
4	0.6268	0.6195	**0.629**
5	**0.844**	0.832	0.84
6	0.6651	**0.681**	0.675
7	**0.341**	0.331	0.34
8	0.98	0.979	**0.982**
9	**0.657**	0.6332	0.64
10	**1.00**	**1.00**	**1.00**
11	0.881	**1.00**	**1.00**
12	0.97	**1.00**	**1.00**
13	**1.00**	**1.00**	**1.00**
OA	0.838	0.841	**0.846**
AA	0.824	0.834	**0.837**
KC	0.823	0.828	**0.831**

TABLE IX
TIME CONSUMPTION COMPARISON ON DISJOINT PU DATASET

Model	Training Per epoch	Inference Time Per Sample	Model Size (M)
1-D CNN	**3.2 s**	**0.028 ms**	**234**
2-D CNN	14.2 s	0.27 ms	1012
SSRN	202 s	0.11 ms	829
A^2S^2k-ResNet	204 s	0.180 ms	846
S^2RGANet	210 s	0.221 ms	754
RIAN	9.4 s	0.106 ms	237
RepSSRN	212 s	0.110 ms	850
Proposed	75 s	0.193 ms	379

are more diverse data points across a larger area, compared to PU where the data points are closer and from a less diverse area, relatively rendering it a more challenging dataset. From these observations, it can be concluded that, while the RepSSRN model performs well in a randomly sampled PU dataset owing to its high parameters, that alone is not sufficient to ensure generalized improved performance across other datasets.

3) Ablation Study: To examine the effects of the various components in the method, ablation studies are conducted.

a) Effect of optimization methods: Results of the ablation study are provided in Table VI to observe the effectiveness of the proposed modules and training methods. Without any additional modules, the proposed 3D-S2FN module performs well. In the subsequent experiment, the GAFP module is added parallelly and the feature vectors from both 3D-S2FN and GAFP are concatenated and passed through a fully connected block. It is to be noted that the GRAM is not utilized here, as the map is not being supervised using a binary class map prediction scheme. Incorporating the GAFP module in this manner boosts both the OA and AA scores. However, training the models with both the classification loss and binary class map loss boosts the AA score by a large margin. This can be attributed to the increased generalizability of the proposed GAFP module across classes as well as the effectiveness of the GRAM method with binary map supervision (denoted by map loss on Table VI).

b) Effect of 3D-S2FN layer depth: To investigate the influence of 3D-S2FN layer depth, the optimal method identified in the optimization ablation study (utilizing 3D-S2FN with GAFP, both classification and map loss) is selected, and variations were made in the number of residual EPSA modules (referred to as layer depth) within the 3D-S2FN module. As shown in Table VI, the four-layer model shows a marginally higher OA score, though its performance is very similar to that of the model with three residual EPSA layers. Conversely, a 3D-S2FN model configured with only two layers exhibits a substantial reduction in performance across both OA and AA metrics. The utilization of three layers yields the highest AA score while retaining the OA score close to the highest achievable result, which makes this the optimal model considering the number of parameters. Experimental iterations are extended up to a layer depth of four, as it may be inferred that, further augmentation of layer count would yield insignificant performance increases while increasing the model parameters significantly.

c) Effect of central and input HSI cube size variation: The effect of the input HSI cube spatial size and the central HSI cube spatial size and their interactions are examined in Table VII. From the table, it can be observed that, although having the input and central HSI cube spatial resolution at 9×9 and 5×5 pixels, respectively, yields the highest OA and KC scores, this increase is offset by the low AA scores. A model with 7×7 input resolution and 3×3 central resolution attains the highest AA score while maintaining decent OA and KC scores. Despite its comparatively lower OA and KC scores, this configuration is ultimately selected as the proposed method, primarily due to its superior performance in AA, surpassing the capabilities of existing SOTA models. This increase in AA can be attributed to the central pixel features being prioritized more in smaller central HSI cubes, which can help the model differentiate the classes.

d) Effect of 3D-S2FN and GAFP: To examine the effects of the distinct modules in the proposed method, the individual results of the 3D-S2FN and GAFP submodules are examined. In this analysis, the feature vectors generated by these respective submodules are fed to classifiers, effectively

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TABLE X
COMPARISON WITH RECENT WORKS ON IP SCENE

	OA	AA	KC
WFCG [16]	0.906	0.878	0.895
GraphGST [41]	0.97	**0.984**	0.966
MRCAGCFN [50]	0.969	**0.984**	0.965
S^2GFormer [51]	0.937	0.965	0.926
Proposed	**0.981**	0.972	**0.984**

establishing them as independent networks for HSI classification. In Table VIII, the best-performing network among the 3D-S2FN and GAFP on a specific class (class-wise accuracy) is highlighted in green to indicate the comparative performance between these two networks. As can be observed from the table, though the GAFP network performs better in predicting some specific instances, it falls short in other scenarios. Conversely, the 3D-S2FN consistently achieves high overall scores in terms of OA, AA, and KC, despite occasional difficulties in classifying certain categories. But when they are combined in the proposed method along with their interactions through the GRAM mask, the results are higher for most classes and the overall OA, AA, and KC scores surpass the performances of individual 3D-S2FN and GAFP networks. This substantiates the efficacy of incorporating both submodules within the framework of the proposed methodology, along with the effectiveness of GRAM.

4) Time Comparison: The time taken by the model is compared against existing models in Table IX. This analysis is undertaken by recording the training duration per epoch and the inference time per sample on the PU dataset. Although the 1-D-CNN [28] is the fastest, its performance is suboptimal compared to other existing models. Comparatively, RIAN [30] performs better while maintaining decent scores. The training and inference times associated with the proposed method are positioned at an intermediate level. While this model does not claim the distinction of being the fastest, it is noteworthy that certain models require significantly more time for both training and inference processes. Additionally, the improved classification results across the four datasets highlight the capability of the proposed method.

5) Comparison With Recent Work: To validate the proposed SSGRAM method in the context of recent advancements in HSI classification, additional comparisons are conducted with SOTA methods, with an emphasis on methods that incorporate GNNs in their architecture. In WFCG [16], CNN and GNN outputs are merged using a weighted addition strategy. MRCAGCFN [50] is selected for its innovative fusion of adaptive GCNs and CNN kernels with random shapes for HSI classification. S^2Former [51] is included due to its novel spectral–spatial graphormer network architecture. GraphGST [41] is chosen for its incorporation of graph structural embeddings within the vision transformer framework. Table X presents a comparative analysis with the IP dataset, where the proposed SSGRAM approach demonstrates competitive or superior performance against these SOTA methods. Specifically, an OA of 0.981 is achieved, surpassing GraphGST by 1.1%, MRCAGCFN by 1.2%, and S^2GFormer by 4.4%. Similar improvements are observed in the KC metrics, where SSGRAM outperforms most recent methods. Although the proposed method is slightly outperformed by GraphGST and MRCAGCFN on AA, it still provides competitive performance. These results substantiate that the proposed method effectively addresses the limitations of existing approaches by integrating both CNN and GNN architectures through the GRAM mechanism.

V. CONCLUSION

In this article, a dual-branch deep learning model is proposed for improved HSI classification. The proposed 3D-S2FN module selectively prioritizes features processed across different dimensions for improved feature processing. Ablation studies demonstrate that this module provides better classification results than the existing methods. Additionally, to mitigate neighborhood bias, the proposed GAFP module compares the center pixel with the surrounding pixels and generates an attention mask and feature based on the comparison. The experimental results indicate that the supervised attention mask helps the model perform better, especially in cases where the surrounding pixels belong to different classes. To the best of the authors' knowledge, the incorporation of attention maps from a GNN-based network to enhance features in 3-D-CNN-based models is a novel contribution that has not been explored in the existing literature. The proposed SSGRAM method outperforms the existing SOTA methods across disjoint sets in four datasets, with an average increase of 1.1%, 1.5%, and 1.5% in OA, AA, and KC scores, respectively. One of the possible drawbacks of this method is the relatively high inference time, so our future works will explore methods to minimize it while retaining performance.

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